Diamond: A Sequence-to-Sequence Model for Speech Restoration via an Autoregressive RQ-Transformer over Neural Audio Codec Tokens

Technical Report

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Abstract. We present **Diamond**, a **sequence-to-sequence** speech-restoration model that turns **degraded audio into studio (or near-studio) quality** at 44.1 kHz. The degraded mel-spectrogram is processed by a bidirectional Transformer encoder; an autoregressive decoder with cross-attention predicts the tokens of a frozen neural audio codec (DAC, 9 RVQ codebooks). The decoder is an **RQ-Transformer**: a *time-transformer* models the frame sequence, while a compact *depth-transformer* models the codebook axis inside each frame, restoring the causal RVQ chain and distributing the output projection (read-out). Autoregressive models over acoustic tokens have been proposed before [41], but there the codebooks of a frame are read by *parallel* heads from a single vector; to the best of our knowledge, Diamond is the **first speech-restoration model whose read-out is a two-transformer RQ decoder** (a frame-level time-transformer plus a frame-local depth-transformer along the RVQ axis). Diamond is trained **from scratch** (≈166.6M parameters, no pretrained backbone) and reaches **the level of the strongest current open systems** for speech restoration, while remaining **the best from-scratch restorer in its class**.

Index Terms — speech restoration, sequence-to-sequence, speech enhancement, neural audio codec, RVQ, autoregressive models, RQ-Transformer, DNSMOS.

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1. Introduction

Diamond addresses **speech restoration**: recovering **studio (or near-studio)** quality at 44.1 kHz from degraded audio. Real degraded recordings pass through aggressive lossy channels and end up band-limited (low-bitrate codecs cut the spectrum near 8 kHz, removing sibilants and "air"), with codec-smeared transients, background noise, and hard clipping. The recording nevertheless remains **intelligible but low-quality**. Such audio is typical of podcast datasets, interviews, and archival or user-generated recordings that have passed through various lossy codecs (MP3, AAC, Opus) and transmission channels; the method is not tied to any particular source and improves a range of degraded corpora.

Such recordings **dominate** real speech corpora, they cannot be re-recorded, and as-is they are **unusable as studio targets** for training — the input dirt is reproduced by the model at the output. Restoration ("dataset cleansing") turns large volumes of otherwise useless data into trainable data; this is exactly the task addressed by industrial systems (Miipher 2 [19]).

Restoration is not denoising. The distinction matters because it determines the class of model. In denoising, the observation follows an additive–convolutional model $y = (x * h) + n$: the phonetic and harmonic structure is *fully present* in y, merely contaminated. The task reduces to estimating a mask, $\hat{X}(f,\tau) = M(f,\tau)\,Y(f,\tau)$, and deterministic convolutional networks solve it well. In restoration part of the signal is **physically absent**: the codec cut the band above

8 kHz, clipping cut the peaks, a lost packet zeroed the frame. No mask can return what the observation does not contain. The missing content must be **generated** from the context of the surviving formants — restoration is therefore a generative problem, not a filtering one. Hence our choice of a generative seq2seq model (Sec. 2.2).

A further difficulty is that a restorer must remove the degradation **without changing the words**. A generative model may sound clean (high DNSMOS) yet smear or substitute content — therefore we measure content preservation (CER) as a separate, mandatory axis.

Hypothesis. A compact sequence-to-sequence model — mel → Transformer encoder → autoregressive decoder over the tokens of a frozen DAC [1] — can restore degraded speech to near-studio quality **without any pretrained backbone** and reach the level of the strongest open systems. Diamond takes the **codec-token** approach: a frozen DAC (44.1 kHz, 9-codebook RVQ) defines a discrete target space at 86 Hz; the model predicts these tokens from the degraded input and hands them to the frozen DAC decoder. This separates "what to say and how it should sound" (learned) from "how to render tokens into audio" (a fixed, high-quality decoder).

The path from hypothesis to a working model was **not free**: deep training from scratch over codec tokens is unstable (encoder layer-0 runaway, read-out inflation, gradient-spike storms). We document each of these problems and its solution in Sec. 3. The hypothesis **held**: Diamond reaches the level of the strongest current open restorers and is the best *from-scratch* restorer in its class (Sec. 6).

Autoregressive models over RVQ tokens (MusicGen [2]) and the two-transformer RQ-Transformer scheme (Lee et al. [3]; Moshi [4]) have been studied in music generation, TTS, and dialogue; their application to **speech restoration** is new.

Contributions. (1) A **two-transformer RQ decoder** over neural codec tokens applied to speech restoration: not parallel per-codebook heads, as in prior AR models over codec tokens [41], but a frame-local depth-transformer that restores RVQ causality and distributes the read-out. The model is trained from scratch (Sec. 2). (2) Diagnosis and resolution of the key instabilities: layer-0 runaway, read-out inflation, the exposure-bias cascade — with mechanistic explanations grounded in the literature (Sec. 3). (3) A data and augmentation recipe: 681.5 h with 28% native wideband content (Sec. 4). (4) An honest two-axis evaluation (DNSMOS-P.835 + CER) against open baselines on real bad recordings (Sec. 6).

2. Our method

2.1. Overview

Before describing the blocks, we explain the two decisions that shaped the whole architecture: **why** the model is a sequence-to-sequence Transformer rather than a diffusion model (Sec. 2.2), and **why** its decoder predicts discrete tokens rather than a spectrum or a waveform (Sec. 2.3). Both follow from a single requirement: remove a *mixture* of unknown degradations in one pass, without damaging the content.

2.2. Why a Transformer rather than diffusion

In short. A diffusion model is conditioned on *one* degradation operator and is brittle with respect to it; our recordings carry a mixture of unknown distortions. What is needed is an architecture in which a single conditional module sees the whole signal at once and removes the distortions jointly — that is a seq2seq Transformer with cross-attention. Below is how we arrived there.

Our first formulation was a diffusion one. The starting point was **NU-Wave** [31], a diffusion model for audio super-resolution; the same line continues in AudioSR [32]. The model reconstructed the missing high frequencies confidently, but it repaired **exactly the degradation it was formulated for**: it is conditioned on a band-limited signal and fills in the missing band.

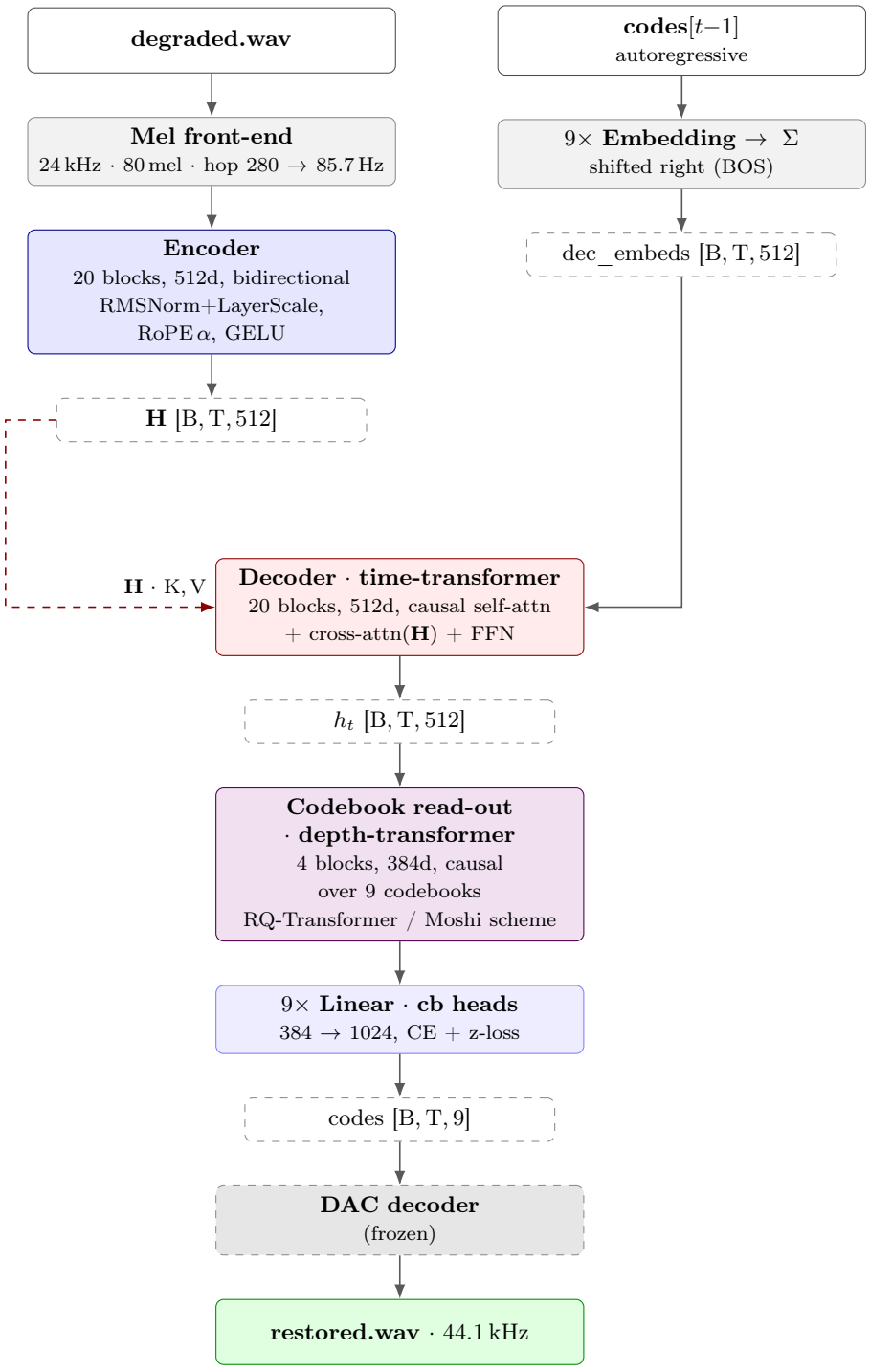


Figure 1: Diamond architecture (dataflow). Degraded audio → log-mel → a bidirectional **encoder** produces the context **H**; an autoregressive **decoder** (time-transformer) with cross-attention to **H** models the frame sequence and emits one vector h_t per frame; the **codebook read-out** (depth-transformer, N×4) generates the 9 residual RVQ codes of the frame sequentially (RVQ causality restored, RQ-Transformer/Moshi scheme); the frozen **DAC decoder** synthesizes the 44.1 kHz waveform. Dashed boxes are intermediate tensors. Only the encoder, decoder, and read-out are trained (≈166.6M); DAC is frozen.

Noise, codec-smeared transients and clipping simply do not exist in that formulation. Neighbouring work states the same limitation explicitly: BABE [33] notes that it accounts for *bandwidth limitation and additive noise only*, whereas real recordings also suffer from coloration, distortion and pitch variation, which its model leaves out.

Our working hypothesis at the time was this: a generative model restores only from the *narrow distribution of degradations* it was trained on, and falls apart as soon as the input becomes diverse. It turned out to be *almost* right — and we refine it below. It is wrong to charge this limitation to diffusion as a model class; it belongs to the conditional formulation, in which the distribution of degradations is baked into training.

The mechanism: why a single degradation. Restoration in the diffusion formulation is an **inverse problem**. The observation is tied to the original by a degradation operator $\mathcal{A}$: $x = \mathcal{A}(y)$, and the model samples from the posterior $p(y \mid x)$. There are two routes, and both hinge on $\mathcal{A}$. *Conditional training* (NU-Wave, AudioSR) does not need $\mathcal{A}$ explicitly — but it **bakes it into the weights** through the training distribution: the network learns the posterior for precisely the $\mathcal{A}$ that generated the pairs. *Guided sampling* with an unconditional prior (CQT-Diff [37]) is free of specific pairs — but it requires $\mathcal{A}$ **explicitly and differentiably**, in order to pull the sample towards agreement with the observation at every step. Hence a separate line of work on the "blind" case: BABE [33] is devoted to *estimating* the unknown filter — a problem that exists only because the scheme does not work without $\mathcal{A}$.

Our $\mathcal{A}$ satisfies none of these requirements. It is **unknown** (the codec and its settings are not recorded in the file); **compound** — codec, noise, clipping and packet loss act jointly and do not commute; **non-differentiable and non-invertible** — a codec discards information irreversibly and adds quantization, and a lost packet cannot be recovered by anything; and it **varies from recording to recording**. Estimating $\mathcal{A}$ per file, as the blind formulation does for a single filter, is unrealistic for a mixture of distortions.

Worse, such models are **brittle with respect to the assumed degradation operator**. Moving NU-Wave 2 from the low-pass filter it was trained on to a filter of a different type — at the same cutoff — drops quality from 19.17 to 10.83 dB SI-SNR [34]. Our recordings are podcasts and interviews that have passed through unknown codecs with unknown parameters; the assumption "the degradation operator is known exactly" is unavailable to us.

We tried to widen coverage with a **cascade**: a diffusion super-resolution stage followed by an adversarial refinement stage. The cascade added complexity — two models, two training regimes, error accumulation between stages — but **did not widen the set of covered artifacts**: each stage still solved its own narrow task. We report this as a negative result; it was informative, because it showed where the limitation actually lies.

The limitation is not diffusion as such. This must be stated honestly, and here our initial hypothesis needs a correction: diffusion models do handle mixtures of degradations (UNIVERSE [36] trains on 55 distortions simultaneously) and multi-task use without retraining (the unconditional CQT-Diff prior [37] is applied zero-shot to bandwidth extension, inpainting and declipping). So it is not the generative nature of the model, nor diffusion. The limitation lies in the **formulation**: one conditional model per degradation operator. "It fails under diversity" is an observation about the *support of the training distribution* of $\mathcal{A}$, not about a family of models. The right response is to remove $\mathcal{A}$ from the formulation, not to swap diffusion for something else.

Two systemic costs we also did not want to inherit. First, **speed**: generation requires dozens of sampling iterations per recording, an order of magnitude more costly than a single predictor pass. Second, **content fidelity**: a survey of diffusion methods for audio restoration [35] explicitly notes a tendency to *hallucinate* at low input SNR — inpainting speech into noise-only regions, breathing artifacts, and the introduction of phonetic confusion. This too follows from the structure of the posterior: when the observation barely constrains the solution (heavy noise), or constrains it *wrongly* (a mismatched $\mathcal{A}$), the likelihood term weakens and the sample is decided by the **prior** — the model of how speech sounds in general. That prior then paints in plausible

speech that was never in the recording. The same effect shows up in numbers: Resemble Enhance [20], whose enhancer is a conditional flow-matching model, reaches a high DNSMOS at a content accuracy of roughly 47% [5]. This is precisely why we measure CER as a mandatory second axis (Sec. 6).

Hence the architectural requirement: a formulation in which **one** conditional module sees the **entire** degraded signal and maps it to a target representation, removing the artifacts *jointly*, in a single pass, without assumptions about the operator. That is precisely sequence-to-sequence: a bidirectional encoder builds the context of the whole recording, and a decoder with cross-attention generates the clean version. The Transformer is not an end in itself but a consequence: cross-attention gives the decoder arbitrary access to the whole input rather than a local window. This is necessary because the artifacts are **non-local** — codec smearing and packet loss are spread over time, and the cause of a corrupted frame may lie far from the frame itself.

2.3. From a continuous output to discrete tokens

The second decision is why the decoder predicts **discrete tokens** rather than a continuous spectrum or waveform.

In short. Predicting a continuous target is *regression*, and regression averages all plausible variants, muffling the sound. Predicting a token from a finite vocabulary is *classification*, and instead of averaging it *selects* one variant. The decoder therefore emits a code index rather than a spectrum: at each frame it receives its hidden vector h_t and **selects** one code from every RVQ codebook; turning codes into sound is the job of the frozen DAC decoder.

Where this decision came from. It grew out of the same cascade experiment. The adversarial stage was there for exactly one purpose — to bring back the detail that the preceding stage had smoothed away. In other words, the GAN was treating the *symptom* of averaging rather than its cause, and we paid for that with a second model and unstable training. We later saw the same pattern in other systems (below). Hence the question that shaped the architecture: can the *cause* be removed, so that no adversarial stage is needed at all? The answer was suggested by the line of audio language models [38, 2]: if the target is **discrete**, there is nothing to average — the model does not blend variants, it selects one. We carried that formulation over to restoration, where the conditioning signal is the degraded audio.

Regression to a continuous target under an L_1/L_2 loss converges to the **conditional mean** over all plausible realizations. For restoration this is fatal: the fine structure (sibilants, plosive onsets, phase) is **multimodal**, and its mean is a smoothed, muffled signal. The effect has been known in speech synthesis since the work on global variance [40]. This is exactly why regression systems have to bolt on an adversarial stage: RE-USE [5] trains a regression model to convergence, freezes it, and puts a GAN *on top* to repaint the *over-smoothed* regions. Realism returns at the price of a second model and unstable adversarial training.

Discretization removes the problem at its root. A frozen RVQ codec defines a finite vocabulary, and restoration becomes **classification**: the model *selects a mode* instead of averaging modes. As WaveNet [39] puts it, a categorical distribution is more flexible because it makes no assumptions about the shape of the distribution. Hence:

- **No over-smoothing.** Multimodality is represented explicitly: a softmax over 1024 codes holds several plausible options, and argmax commits to one of them whole.
- **A stable objective.** Cross-entropy is well conditioned; no discriminator, no noise schedule, no adversarial training.
- **The frozen decoder as a prior.** DAC is trained to map codes to *natural* audio, so even imprecisely predicted tokens render into a plausible signal rather than a spectral artifact: the output lies on the manifold of real speech.

- **Separation of concerns.** "What should sound" (learned) is separated from "how to render it" (a fixed, high-quality decoder) — hence the modularity in the codec.
- **A tractable length.** The target is 86 Hz × 9 codebooks instead of 44 100 samples per second, which makes autoregression computationally realistic.

The framing of "audio as a sequence of discrete tokens for a language model" goes back to AudioLM [38] and MusicGen [2]; other work carries it over to restoration as well [41]. Our contribution is not the transfer itself but the design of the read-out (Sec. 2.8).

The honest price of discretization, which we do pay: (i) the ceiling of the system is **bounded by DAC's own reconstruction** — even with perfectly predicted tokens the output cannot exceed what the codec can rebuild from them; (ii) autoregression is serial and slower than non-AR predictors; (iii) exposure bias appears (Sec. 3); (iv) prediction accuracy **degrades with the RVQ codebook index** — the fine codebooks carry a residual that is ever less predictable from the degraded input (Fig. 2).

Why the decoder is conditioned on clean audio. A training pair consists of a *degraded* input x and the *clean* original y; they are synchronous by construction, because x is produced by augmenting that same y. The frozen DAC is applied **to y only**: it encodes the clean audio into codes $\mathbf{c} = \mathrm{DAC.enc}(y)$, and those codes are the training target. The degraded input **never passes through the codec**: it enters the model as a log-mel and goes on to the encoder. The asymmetry is deliberate: *the input is what there is; the output is what there should be.*

Tokenizing the input as well would be a mistake, for two reasons: one about content, one about cost.

Content. A codec is a faithful compressor: the codes of a degraded signal would carefully encode *the artifacts themselves*, and a 44.1 kHz codec fed a band-limited input would spend part of its codebooks on the empty upper band and on noise. A mel-spectrogram, by contrast, hands the encoder a continuous representation with no input-side quantization loss, and the encoder itself decides what is signal and what is artifact.

Cost. The input is **synthesized online** on every access (Sec. 4), so its representation must be recomputed at every training step. A mel is a single STFT: 80 channels at 85.7 Hz, about $6.9{\cdot}10^3$ numbers per second against 24 000 waveform samples. Tokenizing the input would mean running *another* neural encoder (DAC at 44.1 kHz) on every sample of every epoch — extra memory, extra time, extra resampling. Discreteness is needed **where we generate**, and not where we merely condition.

This formulation also pays off in practice: the targets are computed **offline** once (a single pass of the DAC encoder over the corpus), and the codec is absent from the training loop altogether — neither its memory nor its gradients; DAC stays frozen (Sec. 4). The price is exactly the one any autoregression with teacher forcing pays: during training the decoder sees *clean* codes $\mathbf{c}_{<t}$ in its context, whereas at inference it sees its own predictions. This is exposure bias, and we treat it separately (Sec. 3).

The decision to "predict tokens" raises the next question: *whose* tokens. The vocabulary is set by the codec, and with it the bandwidth ceiling, the temporal resolution, and the quality ceiling of the whole system.

Why DAC as the target codec. The task is to restore the **full studio band** (0–22 kHz), so the codec must deliver that band. DAC [1] is a 44.1 kHz RVQ codec with high reconstruction fidelity, and this determined the choice:

- **44.1 kHz output → a 22 kHz bandwidth ceiling.** Codecs at 22–24 kHz (NanoCodec [21], Mimi [4]) have a bandwidth ceiling of 11–12 kHz — they physically cannot bring back sibilants and transients above it, which is critical for restoring studio quality.
- **An 86 Hz frame rate → an 11.6 ms step.** This suffices to resolve plosives (/t/, /k/, 5–15 ms long); codecs at 12.5 Hz (an 80 ms step) smear them.

- **Discrete tokens → a classification problem.** The target space is discrete, and restoration reduces to stable cross-entropy training (rather than spectral regression). A frozen high-quality decoder separates token prediction from audio rendering.
- **Open weights, battle-tested, modular.** Our `FrozenAudioCodec` interface allows swapping the codec with a minimal diff should a better 44.1 kHz option appear.

DAC is frozen at every stage: it provides the training targets (encoding clean audio → codes) and the final synthesis (codes → waveform). Only the encoder and the two transformers are trained.

2.4. The closest prior work, and where we diverge from it

Closest to our formulation is the **Speech Enhancement Transducer** (SET) [41], an autoregressive model over acoustic tokens. On three key decisions we diverge from it in the *opposite* direction, and the divergence is substantive.

(1) Input representation. SET feeds the model *tokens of the noisy signal*; we feed a log-mel and apply the codec to the clean target only. A codec faithfully encodes the artifacts as well (see above).

(2) Backbone. SET borrows the transducer from speech recognition (encoder + predictor + joiner) and **deliberately drops cross-attention**: its authors treat the task as "many-to-many", where every input token has a corresponding output token, and therefore consider a full encoder–decoder an inefficient use of capacity, since no alignment has to be learned. The objection is valid *for their setting*: their input and output are tokens of the same codec, so the correspondence is token-wise. Ours is different: the input is a log-mel of a degraded 24 kHz signal, the output are codes of a clean 44.1 kHz signal. Frames are aligned in time but **not** in content: the information required is simply absent from the corresponding input frame (the upper band is gone). Cross-attention serves not alignment but access to *neighbouring* frames — a sibilant is restored from context, not from a frame that does not contain it. This follows directly from the non-locality of the artifacts (Sec. 2.2).

(3) Read-out. SET predicts the codebooks of a frame in parallel — and reports the price itself: moving from teacher forcing to beam search **doubles** the content error, because it "increas[es] the likelihood of at least one codebook prediction being incorrect"; a separate refinement step was needed to close part of the gap. We instead distribute the read-out across the steps of a depth-transformer (Sec. 2.8).

Finally, the scale of the tasks differs: SET operates at 24 kHz with 4 codebooks (denoising), we operate at 44.1 kHz with 9 codebooks (full-band restoration). The authors of SET conclude that *discrete representations still fall short of continuous ones*. Our results (Sec. 6) show that with different choices for (1)–(3) a token-based model reaches the level of the strongest open systems. We draw no direct comparison — the corpora, the metrics and the task itself differ — but this gives grounds to regard the shortfall as a property of those choices rather than of discreteness as such.

2.5. Mel front-end

The input x is resampled to 24 kHz; we take an 80-channel log-mel spectrogram with hop 280, which yields a frame rate of

$$f_{\text{frame}} = \frac{24000}{280} \approx 85.7 \text{ Hz} \approx \frac{44100}{512} = 86.1 \text{ Hz} = f_{\text{DAC}}, \tag{1}$$

so the two frame rates agree **to within** 0.5%. The match is not exact: 10 s yields 857 mel frames against 861 DAC frames. We do not align them explicitly — cross-attention operates on differing lengths T_{mel} and T_{tok} by construction, and the generation length at inference is set

as $T_{\text{tok}} = \max(1, \lceil N_{44.1}/512 \rceil)$. What matters is that cross-attention is relieved of the temporal "super-resolution" burden it would carry if the rates differed by a factor. Features:

$$\mathbf{X} = \max\big(10 \log_{10} \text{Mel}(|\text{STFT}(x)|^2),\ -80\big), \tag{2}$$

$$\mathbf{X} \leftarrow \tfrac{\mathbf{X}+80}{80} \cdot 2 - 1 \quad (\approx [-1, 1]^{80 \times T}). \tag{3}$$

The cutoff is $f_{\max} = 8000$ Hz: the useful signal of the degraded input lies in 0–8 kHz.

2.6. Encoder

A twenty-layer bidirectional Transformer, d=512. The front-end is two 1-D convolutions (Conv1d($80 \to 512$) and Conv1d($512 \to 512$), kernel 3, stride 1, padding 1) with GELU, followed by `norm_in` (Sec. 3). Stride 1 means **no temporal downsampling** — restoration needs the fine temporal detail (plosive onsets, sibilant energy) that the downsampling of ASR encoders (Whisper [7]) throws away. Blocks are pre-norm with RMSNorm [8],

$$\text{RMSNorm}(x) = \frac{x}{\sqrt{\frac{1}{d}\sum_i x_i^2 + \epsilon}} \odot \gamma, \tag{4}$$

RoPE [9] with a learnable per-layer scale [44] $\theta_i^{(l)} = \alpha^{(l)}\, b^{-2i/d}$ ($\alpha^{(l)} \in [0.1, 2]$), GELU in the feed-forward, and **LayerScale** [10] on every residual branch.

Why GELU rather than ReLU or SwiGLU. ReLU hard-zeroes the negative part → "dead" neurons and a non-smooth gradient; on spectral features (log-mel goes negative) this loses information. SwiGLU adds a *third* projection to the feed-forward (≈+50% parameters/compute); on our already heavy model the overhead is not justified. **GELU** ($x \cdot \Phi(x)$) is smooth and the standard for audio Transformers.

Why RMSNorm rather than LayerNorm. RMSNorm drops centering and bias, normalizing only by the root-mean-square norm → cheaper, fewer parameters, and more robust under mixed precision (we compute the norm internally in fp32). For a 44-layer stack this is critical.

2.7. Decoder (time-transformer, per frame)

A twenty-layer **causal** Transformer, d=512; each layer has causal self-attention (over past frames) and cross-attention (over the full bidirectional encoder output $\mathbf{H}$). It emits one hidden vector per frame, $h_t \in \mathbb{R}^{512}$.

The input at frame t is the sum of the embeddings of the nine codebooks of the *previous* frame; the sequence is shifted right by one step and opened by a BOS token:

$$u_t = \sum_{k=0}^{8} \text{Emb}_k(c_{t-1,k}), \qquad u_1 = \sum_{k=0}^{8} \text{Emb}_k(\text{BOS}), \tag{5}$$

$$h_t = \text{TimeT}\big(u_{\le t},\ \mathbf{H}\big) \quad \text{with a causal mask over } t. \tag{6}$$

The shift guarantees that h_t is conditioned on $\mathbf{c}_{<t}$ and never sees $c_{t,\cdot}$. The within-frame dependency $c_{t,k} \mid c_{t,<k}$ is supplied by the depth-transformer (Sec. 2.8), so the full factorization (7) is realized by two transformers: the time-transformer accounts for $\mathbf{c}_{<t}$ and $\mathbf{X}$, the depth-transformer for $c_{t,<k}$.

2.8. Codebook read-out (depth head of the decoder)

RVQ codes form a causal chain: codebook k refines the residual left by $c_{t,0..k-1}$ of **the same frame** t. The full factorization:

$$p(\mathbf{c} \mid \mathbf{X}) = \prod_{t=1}^{T} \prod_{k=0}^{8} p(c_{t,k} \mid c_{t,<k},\ \mathbf{c}_{<t},\ \mathbf{X})\,. \tag{7}$$

Why one shared projection is not enough. In our earlier experiments the read-out was exactly of this kind — a shared projection in the spirit of MusicGen's delayed pattern [2]. We had to **abandon** it, and here is why. Under it, all nine codebooks are read from a **single** frame hidden vector $h_t \in \mathbb{R}^{512}$ by nine parallel heads $W_k : \mathbb{R}^{512} \to \mathbb{R}^{1024}$, $\hat{p}(c_{t,k}) = \mathrm{softmax}(W_k h_t)$. That single vector must encode the decisions for all nine codebooks: the read-out capacity is **divided by 9** and smeared — the signal goes to the coarse c_0–c_2, while the fine c_6–c_8 get **neither enough representation nor enough gradient** and collapse to the mode (empirically, c_8 accuracy fell to ≈6%; sibilants and transients disappeared). Moreover, the heads are parallel, so $c_{t,k}$ does not see $c_{t,<k}$ of the same frame — RVQ causality is lost. This scheme — parallel heads over the codebooks of one frame — is also used by other AR models over codec tokens: the Speech Enhancement Transducer [41] is autoregressive *in time* but predicts the codebooks of a frame in parallel, and its authors attribute an aggravated exposure bias precisely to this, since it "increas[es] the likelihood of at least one codebook prediction being incorrect". The depth-transformer removes both causes at once: it gives the codebooks separate computation steps (distributing the read-out capacity) and conditions $c_{t,k}$ on $c_{t,<k}$ (restoring causality).

Instead, a compact **depth-transformer** (4 layers, d=384, ≈14M), conditioned on h_t, generates the nine codebooks **sequentially within the frame** with causal self-attention along the codebook axis:

$$\big[h_t,\ \mathrm{emb}(c_{t,0}), \ldots, \mathrm{emb}(c_{t,k-1})\big] \ \longrightarrow\ \hat{p}(c_{t,k}). \tag{8}$$

This restores RVQ causality and **distributes the read-out** across depth steps.

Parameterization. The four depth-transformer layers are **shared across all nine depth steps**: the length-K=9 sequence is processed as a whole, and the identity of a step is carried solely by a learnable codebook position embedding $\mathbf{e}_k \in \mathbb{R}^{384}$ (RoPE is unnecessary here: the depth axis is not time but nine roles from coarse to fine). Only the input embeddings (8×, for $c_{t,0..7}$) and the output heads ($9 \times \mathrm{Linear}(384 \to 1024)$) are per-codebook. The depth axis is vectorized: all $B \times T$ frames run their nine steps as a batch, so the depth-transformer adds only a small fraction of the training compute, with ≈14M parameters against ≈88.7M for the time-transformer.

2.9. Loss

A per-frame weighted cross-entropy with label smoothing ϵ=0.1 over the valid positions Ω:

$$\mathcal{L}_{\mathrm{CE}} = \frac{1}{|\Omega|} \sum_{(t,k)\in\Omega} w_k \, \mathrm{CE}_\epsilon\big(c_{t,k}, \hat{p}_{t,k}\big), \tag{9}$$

with weights $\mathbf{w} = [\,1.5, 0.95, 0.85, 0.75, 0.65, 0.55, 0.5, 0.5, 0.5\,]$. c_0 carries the coarse/semantic content and gets the largest weight; the lifted tail (c_6–c_8 at 0.5) keeps the fine codebooks from collapsing to the mode. We add **z-loss** [11] against logit inflation:

$$\mathcal{L}_z = \lambda_z \, \mathbb{E}\big[(\log \textstyle\sum_v e^{z_v})^2\big], \quad \lambda_z = 10^{-4}, \quad \mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_z. \tag{10}$$

The DAC targets are detached.

3. Training observations and solutions

A deep model over codec tokens, trained from scratch, is unstable. Below are three problems, their **mechanism**, our solution, and the literature confirming that the solution works for the stated reason.

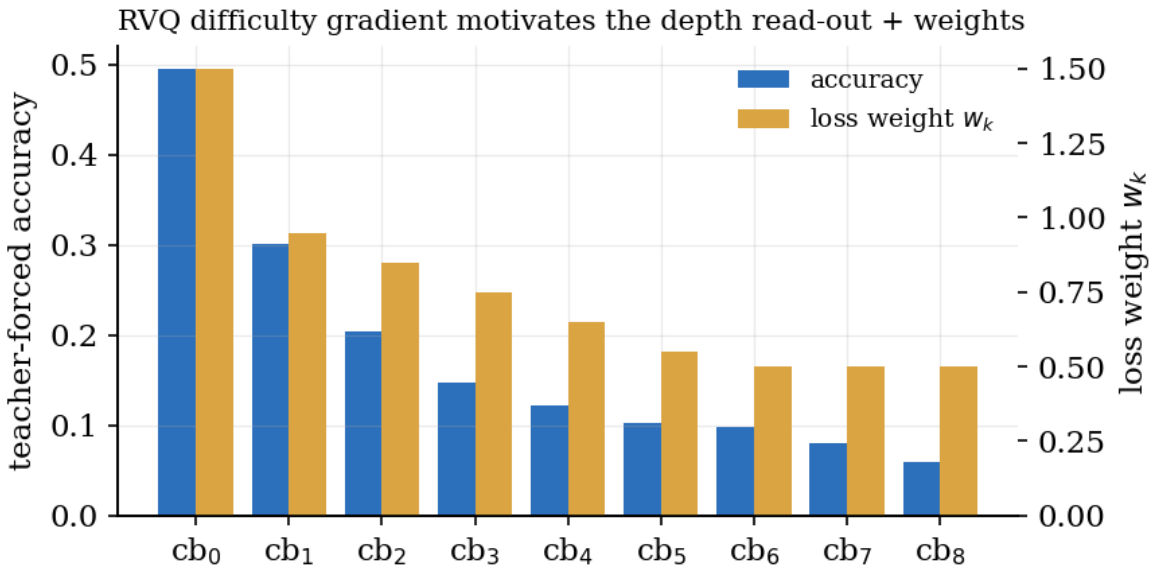


Figure 2: Teacher-forced accuracy across the RVQ codebooks on the **previous architecture** — the one that read all nine books out of a single frame hidden. The fall from the coarse c_0 (≈0.50) to the fine c_8 (≈0.06) is partly unavoidable (a finer book encodes an ever smaller residual and is inherently harder to predict), but the collapse of c_8 to the mode is a read-out defect — and it is what motivated the depth-transformer (Sec. 2.8), with which the tail books recover (c_8: 0.06 → 0.08 by training metrics). The weights w_k lift the semantic c_0 and keep the fine codebooks from collapsing to the mode.

3.1. Encoder layer-0 runaway

Symptom. In early runs the **first encoder layer (L0)** took on a disproportionate share of the representation and **inflated its weight norm several-fold** (the ratio of the L0 norm to that of deep layers grew to 1.5–2×), while the other layers stagnated. The consequence was catastrophic gradient spikes (gn up to 10–90) and a recurring loss regression.

Mechanism. The convolutional front-end ends with an **unbounded GELU**; its high-variance output enters the pre-norm RMSNorm of the first block. **Our hypothesis**: to absorb that variance, the norm's γ and the L0 weights grow — and keep growing through positive feedback. A related phenomenon is **massive activations** in the Transformer residual stream, where individual dimensions acquire huge magnitudes and act as constant biases [15]; there, however, it is described as *benign* and is not tied to weight-norm growth. The positive-feedback mechanism is our own explanation of the instability we observed, not a result from [15].

Solution (and why it works). (1) **norm_in** — an RMSNorm immediately after the convolutional front-end, *before* entering L0. It bounds the input distribution of the first layer at the root, removing the pressure that L0 would otherwise compensate for by growing its weights. (2) **LayerScale** [10] — a learnable diagonal multiplier λ on every residual branch, $x \leftarrow x + \mathrm{diag}(\lambda) f(\mathrm{RMSNorm}(x))$. Each layer tunes its effective contribution with a small scale rather than by inflating weights; this is the very purpose for which LayerScale was introduced in CaiT [10], and it is akin to the residual scaling of DeepNet/DeepNorm [16].

Confirmation. After enabling both, the L0-to-deep-layer norm ratio **decreased monotonically** (≈1.09 → 1.04), the encoder settled into a healthy pre-norm "bell-curve" workload profile, the maximum gn fell from 90+ to ≈5.6, and the regression vanished — exactly as the mechanism predicts.

Table 1: Training data (by source).

Source	Clips	Hours	SR
LibriTTS-R (100/360/500) [13]	230 638	489.4	24 kHz
HiFi-TTS [25]	54 582	77.6	44.1 kHz*
VCTK [26]	49 044	55.4	48 kHz*
EARS [27]	12 132	49.4	48 kHz*
Expresso [28]	5 506	7.3	48 kHz*
Elise (KaniTTS)	1 173	2.4	24 kHz
Total	**353 075**	**681.5**	28% native

*native wideband.

3.2. Read-out inflation

Symptom. In a decoder with a shared read-out, the late layers inflated the output projections, while the fine codebooks c_6–c_8 were undertrained (Sec. 2.8). **Mechanism and solution.** A single 512-dimensional projection split across nine heads dilutes capacity and breaks RVQ causality. The solution is the **depth-transformer** (Sec. 2.8), which distributes the read-out across depth steps; the idea goes back to RQ-Transformer [3] and Moshi [4]. In addition, **z-loss** [11] keeps the logit scale in check.

3.3. Exposure bias and the spike storm

Symptom. Attempting to cure exposure bias with **time-axis** (per-frame) scheduled sampling produced an avalanche of gradient spikes and a loss stall. **Mechanism and solution.** A bad prediction at frame t enters the context of frames $t{+}1, t{+}2, \ldots$ — an **unbounded cascade** along time. We abandoned time-axis scheduled sampling (**it is not used in the release**); the exposure-bias tail is controlled at inference (Sec. 5) and reduced with native data (Sec. 4). **QK-Norm** [12] stabilizes attention, and gradient clipping at norm 1.0 is the last line of defense.

4. Data and training

4.1. Data

Diamond is trained on **681.5 h** of studio speech (**353 075** clips, ≈2.5k speakers). The key property is **28% native wideband content** (≈190 h @ 44.1–48 kHz): the model sees *real* 8–22 kHz content, not merely upsampled 24 kHz targets.

Target selection. Clean targets were selected **by duration only**: clips of 2.5–30 s were kept (shorter ones do not cover the crop window, longer ones inflate memory), after which the manifest was shuffled on disk with a fixed seed. No filtering by perceptual quality — no rejection of targets with low DNSMOS or VQScore — **was performed**: all sources are studio or near-studio corpora. We leave this deliberately as future work: RE-USE [5] reports that filtering targets by VQScore noticeably improves the outcome, and on less clean corpora such a step would be mandatory.

4.2. Augmentation: emulating observed artifacts

The augmentor is not an auxiliary detail of the pipeline but **the second half of the problem statement**. Paired data — degraded and studio versions of the same recording — does not exist in nature: a podcast, an interview, or an archival recording exists only in its damaged form; the

original was never kept. The pairs must therefore be synthesized, by degrading studio targets. And then everything hinges on one question: *which* degradation to apply.

The task we set ourselves: a *natural* degradation, not a synthetic one. Standard augmentation applies effects *of the engineer's choosing*: white noise at some SNR, a filter at some cutoff, a codec at some bitrate. Each parameter is plausible in isolation — but their *combination* is arbitrary, and the result is audio that **occurs nowhere in nature**. We posed the inverse problem: not to invent a degradation but to **reproduce the one that actually happens** to a podcast or an interview — low-bitrate codec compression (often across several passes through different platforms), band truncation, the coloration of a cheap microphone, aggressive dynamic-range compression, room background noise — and to reproduce it *in the combinations and proportions* in which it occurs.

The requirement is strict rather than aesthetic: a model learns to remove exactly the degradations it has seen. If the training input distribution does not match the real one, the model solves a *different* problem flawlessly — and falls apart on an actual recording. This is the same predicament as in Sec. 2.2: the degradation operator $\mathcal{A}$ is unknown to us. We removed it from the architecture; here we settle the same account from the other side — $\mathcal{A}$ cannot be inverted, but one can **sample from a distribution that statistically matches the observed one**. Hence also the criterion we held ourselves to: not "it sounds similar" but **measurable proximity of the feature distributions**.

Inverse calibration. Hence a custom augmentor, built as a fitting problem rather than as a bag of random effects. The target pool is **96 899** real degraded recordings. The similarity metric is built on **eight channel features**: SNR estimate, noise floor, dynamic range, spectral rolloff, the energy ratios in the 3.4–8 kHz and above-8 kHz bands, the high-frequency energy ratio and its temporal instability. The set is *deliberately restricted to the channel*: a wider 17-feature vector carried speaker timbre, and the matching would have tracked the voice rather than the transmission path (the evidence is in Sec. 4.3). The palette holds **5000 DSP parameter combinations**, generated by weighted random sampling and then *measured*: for the median combination, ≈86% of the output features land within the range of real values.

Validation at the matching level. For each sample a random real profile is drawn and k-NN retrieves the nearest palette combination. We checked the strategy on the final palette (5000 combinations, 200 targets): the mean retrieval distance is **5.97**, against **8.75** for a random combination and **7.99** for the median one. The reference point is the *intrinsic* spread of the target corpus, **6.59** — how far two random real recordings lie from each other. Retrieval landed *below* that line, i.e. it hit the physical floor of the problem rather than a limit of the matching. This check operates at the feature level; an end-to-end check on the synthesized audio follows.

Validation at the audio level. The matching metric could be lying — it measures the choice of *parameters*, not the result of applying them. The final check is therefore end-to-end: 100 studio recordings are run through the entire pipeline, features are re-extracted from the *resulting audio*, and compared against real recordings (Fig. 3). The studio sits 3.22 away from the real recordings; our augmentation sits at **0.94**. The gap is closed by **71%**, and the augmented and real clouds **overlap**, while the studio cloud lies apart. That is the proof: we land not "in the direction of degradation" but *inside the distribution itself*.

The DSP chain, in order: downsampling (24/32 kHz) → bass boost → microphone coloration → low-pass (cutoff up to 12 kHz) → dynamic-range compression → **codec** → high-frequency instability (in 25% of combinations) → loudness levelling by soft (tanh) saturation to RMS −16 dB → **real noise transfer** → peak 0.99. The codec parameters come from the same calibration rather than from intuition: MP3 (57%), AAC (23%), OGG (15%), no codec (5%); bitrates 16–128 kbps; **1–3 successive passes** (29% of combinations are double re-encodings, 9% triple) with different encoders and VBR/CBR modes. The noise is **not synthetic**: it consists of real background segments cut from live recordings by the Silero VAD [23]. White noise would be mathematically more honest and practically useless — real background has its own spectral

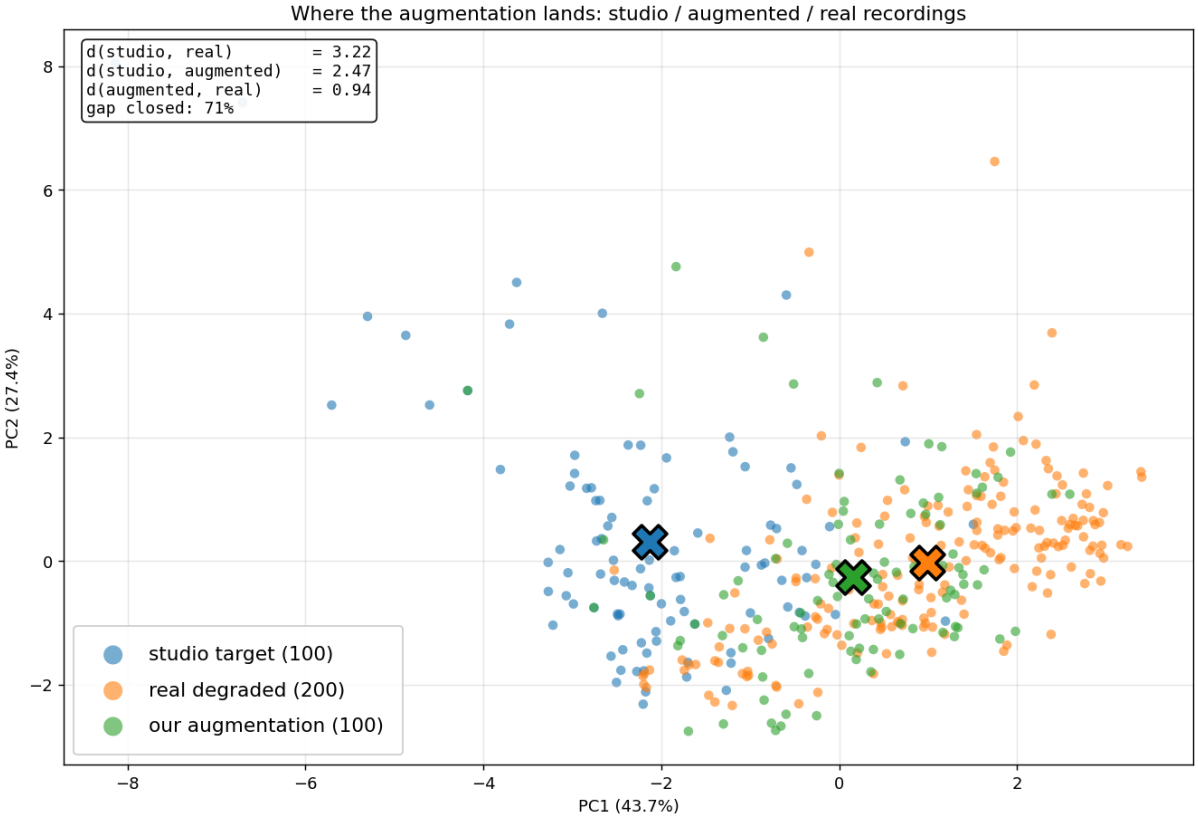


Figure 3: End-to-end validation: the features are extracted from the *synthesized audio*, not from the parameters. Principal components of the feature space; crosses mark the centroids. The studio recordings (blue) form a separate cloud; our augmentation (green) lands **on top of** the real degraded recordings (orange), and its centroid sits 0.94 from the real one against 3.22 for the studio — the gap is closed by 71%.

coloration, and that is what the model must learn to remove.

Guarding against memorization. Pure retrieval draws on a finite set of combinations, and over a long training run the model would start recognizing *specific* DSP signatures instead of generalizing. The sampling is therefore hybrid — 60% k-NN retrieval, 25% a random palette combination, 15% a freshly generated combination with a wider spread — and the continuous parameters are additionally jittered by ±40% within safe bounds. In addition, 5% of samples pass through **undegraded**: the model must also learn not to damage what is already clean.

The robustness track (URGENT-style [24]), each applied with probability 0.15 per sample: **reverberation before the codec** (convolution with real room impulse responses) and **broadband noise after the codec** (SNR uniform in $[0, 20]$ dB). The order is deliberate and matters: echo survives encoding, so reverberation is applied *before* the codec; additive noise is largely erased by the codec, so it is added *after*, otherwise the model would never learn to remove it. Packet loss (0.15), synthetic reverberation (0.10) and hard clipping (0.10) belong here too. As a result the model simultaneously performs dereverberation, denoising, and codec-artifact removal.

All of this runs **online**, inside the DataLoader workers: the targets are tokenized in advance, while the input is synthesized afresh on every access, so the same target is seen under dozens of different degradations across epochs.

4.3. How we got there: stages and rejected hypotheses

The design above is the outcome of four iterations, **three of which failed**. We report them because each refuted a plausible hypothesis, and because without them the final recipe looks like an arbitrary set of choices.

Table 2: Hypotheses tested and their fate. Each row is a separate experiment with a measurable outcome; the decisions we adopted follow directly from the ones we refuted.

Hypothesis	How we tested it	Outcome
Hand-crafted presets will do	feature calibration	no
Take the top-N combinations nearest the center	3-way feature comparison	**failed**
Degradations fall into 4–5 classes	4 clustering criteria	**refuted**
The bass boost is to blame	disable it entirely	no (Δ at noise)
The noise should be broadband	pink noise pre-codec	**36% worse**
The target pool can be taken as is	calibrate on degraded only	target shifted
Uniform parameter sampling	listening A/B	over-augmented
Channel features are speaker-stable	ANOVA / ICC	**yes, ICC 0.60**
Speaker-mean is the better target	listening A/B	no (over-softens)
Probability tweaks can fix a feature	shift three probabilities	no (Δ at noise)
Per-sample retrieval	**vs. intrinsic spread**	**adopted**

v2: "take the combinations nearest the center". The natural move is to rank combinations by distance to the real distribution and keep the best ones. The result: the augmented recordings formed a cloud *of their own*, covering only 56% of the way from studio to target, and ended up **closer to the studio than to the target** (2.51 against 3.77). The cause: ranking by distance to the *center* contracts the sample into a dense core and loses the tails. It also demolished our initial belief that "the target recordings contain no noise": they have a noise floor of ≈ -43 dB, produced by codec quantization, and without it the feature can never match.

Testing the structure: there are no discrete classes. We assumed the degradations fall into 4–5 types, and that covering a representative of each would suffice. The hypothesis looked natural — clustering dutifully produced five groups, and they were even given names. The test (Fig. 4) **refuted** it.

Why clustering alone proves nothing. k-means **always** returns k groups: ask for five and you get five, ask for eight and you get eight. It cannot answer "there is nothing to divide"; it simply cuts the cloud. Our "five types" were therefore a property of the labelling, not of the data. The real question — *does the cut reflect actual structure* — is settled by external criteria, and we applied four independent ones.

The *silhouette* measures how much closer a point is to its own group than to a foreign one. Ours never rises above 0.20 against a 0.25 "decent" threshold: the average point is almost as close to a *foreign* cluster as to its own. And the value is **flat across every k from 2 to 10** — were there a true number of groups k^*, the silhouette would peak there. The *inertia* curve is smooth, with no "elbow": each additional cluster reduces the error by about the same amount — exactly how cutting a *continuous* cloud behaves. The *Davies–Bouldin* index oscillates at noise level with no pronounced minimum. The *gap statistic* compares the data against a **random cloud with no structure at all**: with real clusters it peaks at k^*. Ours grows monotonically and peaks nowhere — that is, **at no k do the data look more clustered than noise**.

Why this is a proof and not a coincidence. Each criterion fails *in its own way*: the silhouette — no separation; the elbow — no saturation; Davies–Bouldin — no minimum of the ratio; the gap — no advantage over the null model. Had a poorly chosen metric been to blame, one criterion would have failed while the others showed structure. **All four failed, each on its**

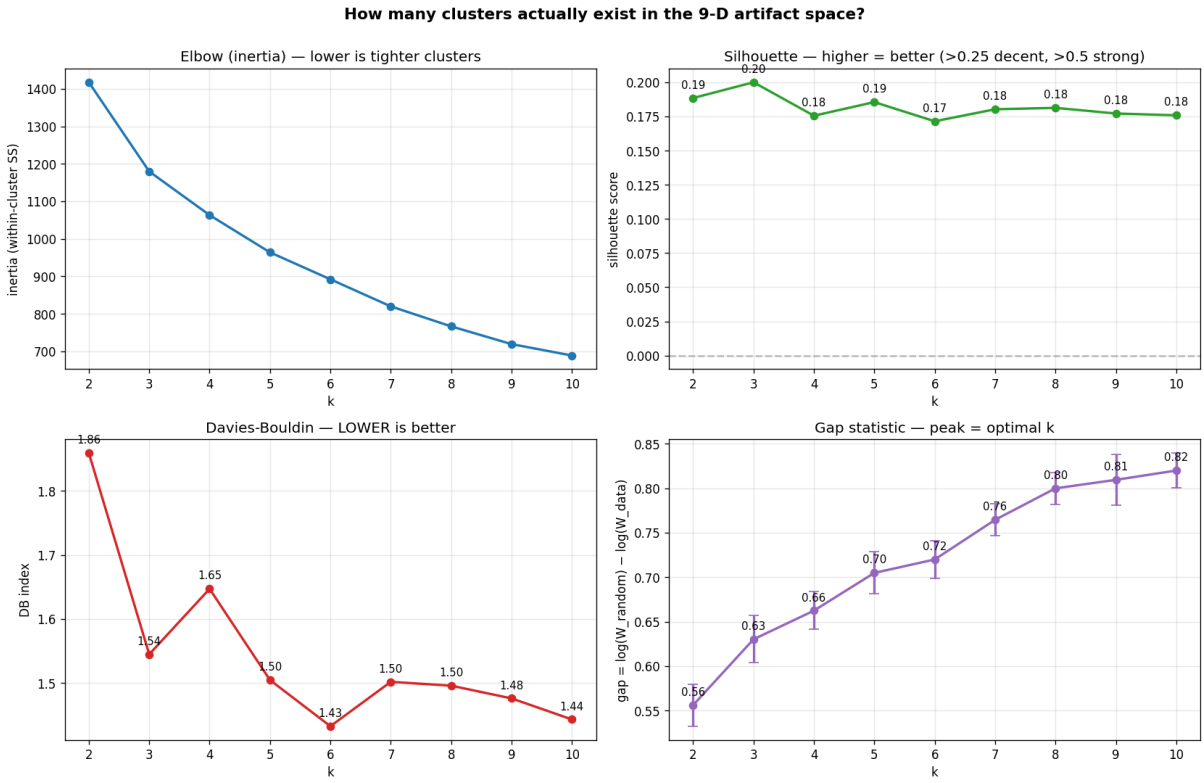


Figure 4: How many clusters actually exist in the artifact space. The silhouette score (top right) never rises above 0.20 for any k, against a 0.25 "decent" threshold. Inertia (top left) is smooth, with no elbow. Davies–Bouldin (bottom left) has no pronounced minimum. The gap statistic (bottom right) grows monotonically, with no optimum. The concordant conclusion: no discrete cluster structure exists — the degradations form a continuum.

own grounds.

We did try to find the structure. The objection "there was simply not enough data" was tested directly: increasing the sample fivefold (200 → 1000) left the silhouette unmoved (0.20), while expanding the feature space from 9 to 106 dimensions made it *worse* (0.13 — the curse of dimensionality). The clusters were not "missed" — they do not exist. One further piece of evidence: at k=5 the largest cluster has a silhouette of 0.12, and **24 of its 93 points have a negative silhouette** — they are physically closer to a foreign cluster than to their own.

What is actually there. Density-based clustering finds **5 density peaks and ≈10% outliers**: the modes *do* exist — some combinations of degradations are more common — but **there are no gaps between them**. Modes are not clusters: a cluster requires a *gap* between groups, whereas a mode is merely a region where the points are denser. It is a mountain range whose peaks can be named but whose borders cannot be drawn: the valleys never descend to sea level.

The practical consequence, which shaped the entire design: what must be covered is the **density in full, tails included**, not a handful of class representatives. Had there been five classes, five profiles would have sufficed. And this *retrospectively explains the failure of v2*: ranking by proximity to the center is precisely the "take representatives" strategy. On a continuum it seizes the dense core and loses the tails — which is what we saw: 56% of the way, and a cloud that settled closer to the studio than to the target.

We audited our own tooling too. The first calibration turned out to contain *no MP3 at all* across its 300 combinations: the random codec draw kept landing on AAC and OGG. We were calibrating against a reality in which MP3 is the dominant format, while the palette did not contain it. Hence a weighted codec choice rather than a uniform one. Later we probed the "obvious culprit" behind the bass skew in the same targeted way — disabling the bass boost

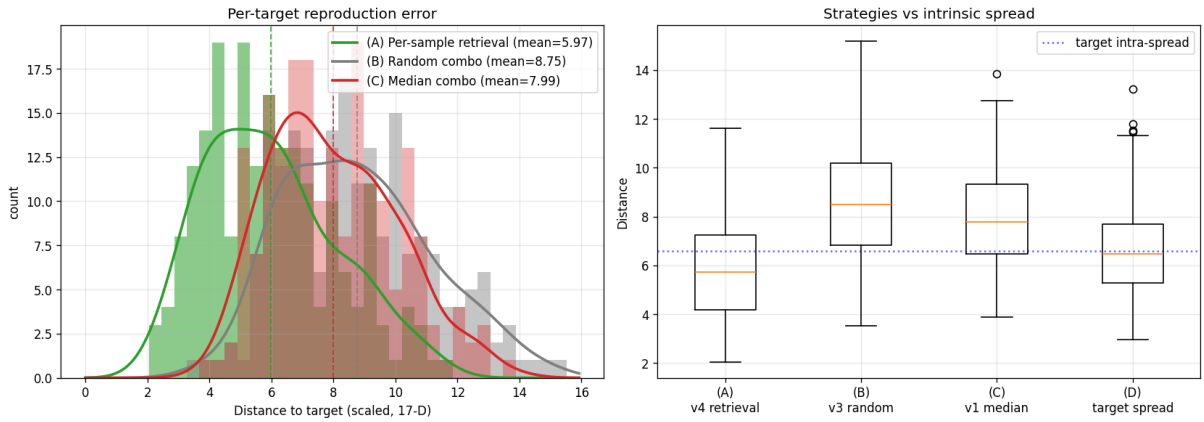


Figure 5: Validation of the per-sample retrieval idea. Left: the distribution of per-target reproduction error for the three strategies (histograms with kernel density estimates; green — retrieval, mean 5.97 — shifted left of random 8.75 and median 7.99). Right: the same strategies against the **intrinsic spread** of the target corpus (D, dotted line at 6.59). The retrieval box (A) sits *at and below* that line: closer to a real recording than real recordings are to each other.

outright; the divergence did not move (Δ within noise). The culprit was not the bass boost but the low-frequency coloration of the noise floor, which inflated the 0–3 kHz band. *The obvious suspect turned out to be innocent* — and that was worth testing rather than assuming.

v3b: "the noise should be broadband". We replaced low-pass-shaped noise with pink noise (physically the more honest choice) and added broadband noise before the codec. The divergence grew by 36%, and the output became *cleaner* than before. The cause proved instructive: **the psychoacoustic model of MP3/AAC deliberately removes noise masked by speech**. Broadband noise added *before* the codec is neatly stripped by the codec. The low-pass-shaped noise survived only because MP3 allocates more bits to the lower bands. Hence the rule baked into the final chain: **noise is added *after* the codec** — otherwise we train the model to remove something that is no longer in its input.

v4: why retrieval rather than random parameters. The key is covariance. Random parameter sampling preserves the *marginal* distribution of each feature taken separately, but destroys **the relations among them**: in real recordings compression and the noise floor are correlated, whereas in a random combination they are not. The result is *chimeras* — artifacts physically impossible in a real transmission path. Per-sample retrieval inherits the entire covariance structure **by construction**: every target *is* a real point with real correlations.

The result (Fig. 5). We compared three strategies on the same set of targets. A random combination (v3) gives a mean reproduction error of 8.75, the median combination (v1) gives 7.99, and per-sample retrieval gives **5.97**. What matters, however, is not the ranking but the **absolute reference**: the fourth column on the right plots the *intrinsic* spread of the target corpus — how far two random real recordings lie from each other on average. It equals **6.59**. Retrieval landed *below* that line: a synthesized degradation is closer to a real recording than two real recordings are to each other. Improving the metric further is meaningless — we have hit the physical floor of the problem, not a limit of the matching.

The target needed a sharper definition than we thought. The first calibration ran against the *whole* corpus. It yielded a target HNR of −9 dB — a catastrophically low harmonicity unreachable by our chain. This proved to be an **averaging artifact**: *clean* recordings had slipped into the pool and dragged the mean. Once the target was narrowed to genuinely degraded material, the true HNR turned out to be −0.8 dB — moderate and attainable. A cheap lesson, but a sharp one: *the distribution you calibrate against matters more than the matching algorithm.*

The metric improved — and the sound got worse. The most unexpected episode. Widening the palette and the parameter bounds raised our gap coverage from 38% to 52% —

progress by every number. The listening verdict was the opposite: *"it over-augmented it"*. The analysis pinned the blame on **uniform sampling over the parameter grid**: bitrates of 16–24 kbps landed in 28% of combinations against ≈5% in reality, triple re-encoding in 33% against isolated real cases, and a telephone-band 3–4 kHz cutoff in 22% against the typical 7–10 kHz. The palette formally covered the distribution but **over-weighted its tails**: the centroid converged while the typical sample sounded harsher than any real recording. Hence **weighted** sampling (bitrates on a Gaussian around 64 kbps, passes 60/30/10), which produced the proportions listed above. This is a fundamental conflict: an aggregate metric pulls toward covering the center, the ear pulls toward typicality. We resolved it in favour of typicality, because the model trains on *samples*, not on centroids.

Channel features are speaker-stable — and that changes the matching metric. The hypothesis: transmission-path artifacts (microphone, room, codec) are physically bound to the session, and should therefore be stable within a speaker. We tested it by analysis of variance over 1500 recordings from 50 speakers: the channel features have **ICC** = 0.60 (60% of their variance is explained by speaker), and `mic_freq_irregularity` reaches **0.96** — a microphone is physically bound to a person. The hypothesis held — but the by-product mattered more: the *timbre* features have ICC = 0.67. That is, matching on the full feature vector would match **the channel *together with* a foreign speaker's voice**, baking that timbre into the augmentation. This is what **quantitatively** justified moving to the *channel-only* feature subset — exactly the one in use now.

Speaker averaging, however, did not work. The logical next step: if the channel is stable within a speaker, use the *mean* over their recordings as the target — phonetic noise averages out, the path signature remains. The metric dropped only slightly (72% against 80%), but the sound "over-softened", drifting back toward the studio. The reason: averaging suppresses not only the phonetics but the intensity of the degradation itself. **The target determines the character of the augmentation** — and per-sample stayed in production.

What you do not measure, you do not match. A systematic diagnostic over 105 features gave a clean cut: all 8 features *inside* the matching metric agreed perfectly (Wasserstein distance < 0.34), while the features *outside* it diverged by 1.5–3.7 SD. Retrieval works flawlessly — precisely in the dimensions it was given. Everything else is left to the DSP chain. That is the boundary of the method, and it is worth knowing.

Probability tweaks do not cure an architectural problem. We tried to close the divergences of the previous paragraph by adjusting the probabilities of the operations (compression 75%→90%, and so on). The shift amounted to ±0.1 SD — within noise. The conclusion saved a great deal of time later: *if a feature misses by 2–4 SD, probabilities will not fix it* — what is needed is a different operation in the chain, or a different feature in the metric.

Acceptance gate: a blind test. None of the "by ear" verdicts above are free impressions — they were made in a **blind test**. The protocol: 20 synthesized and 10 real degraded recordings are placed in one folder under anonymized names (`sample_01..30`) in random order; the file-to-origin mapping sits in a sealed key opened only after the listener has judged every file. The acceptance criterion is **indistinguishability**: if the synthesized recordings cannot be reliably separated from the real ones, the augmentor is good enough; if they are picked out confidently, there is an audible artifact the metrics missed and it must be hunted down. One obvious confound is closed off deliberately: the studio source is **multi-speaker**, because with a single speaker the listener memorizes the voice and starts identifying synthetic samples by *timbre* rather than by channel quality.

The blind test turned out to be the primary detector: both major chain fixes (removing post-codec noise, capping high-frequency instability) and the switch away from uniform grid sampling are *its* verdicts, not the metrics' — in those same episodes the metrics reported *improvement*. It was run iteratively after every fix (seven sessions in a single day at the final stage) and acted as a gate: as long as the synthetic samples were being identified, the fixes continued.

An honest limit: the test was used as a **qualitative gate** — verdicts and specific problem samples were recorded, but the *identification rate* (or d') was not. We therefore report no indistinguishability figure and claim no statistical significance; we claim only that the test was genuinely blind and that every chain fix was dictated by it.

Listening diagnostics: inverse correlation as a signal. A separate methodological lesson. In the blind test the worst-sounding samples were those with a *close* retrieval match, while the best was the one with the *farthest*. The correlation is **inverse** — and that is precisely what needed to be understood: had the retrieval been at fault, the correlation would have been direct. An inverse one means the noise is generated by **the DSP chain itself**, not by the choice of profile. Two phantom artifacts were found this way: post-codec broadband noise (which produced a hiss absent from real recordings) — switched off *entirely* — and an overstated high-frequency instability, whose amplitude we capped at its lowest setting (the operation remains in the chain, in 25% of combinations and at its weakest level).

Peak normalization. *All* thousand real recordings we inspected have a peak of exactly 1.000 — brick-wall limiting. Our chain (compression plus codec) dropped the peak to 0.2–0.3, leaving the output quiet. Normalization to 0.99 is therefore applied *always*, not only when the peak exceeds it.

The flip side of retrieval — and the defense against it. An exact distributional match created a new risk: the palette is finite, and the model may **memorize the signatures** of specific DSP combinations and the fingerprint of a single encoder instead of generalizing. Hence the measures listed above: hybrid 60/25/15 sampling, ±40% jitter, a pool of several encoders (each with its own bit-allocation scheme), and the transfer of *real* noise instead of synthetic. The most dangerous case of overfitting, however, lay not on the input side but on the output: training on a *single* speaker yields a model that transfers foreign speech onto that speaker's voice — which is cloning, not restoration. The only cure is multi-speaker studio targets (Sec. 4, ≈2.5k speakers).

4.4. Training recipe

Only the encoder and the two transformers are trained (DAC is frozen): encoder 63.9M, time-transformer 88.7M, depth-transformer 14.0M — **166.6M in total**. The optimizer is AdamW ($\beta = (0.9, 0.95)$), bf16-mixed, gradient clipping at norm 1.0; the full configuration is in Appendix A. The model is trained **from scratch** — without any external pretrained backbone (unlike systems built on SSL extractors such as w2v-BERT [22]). The released checkpoint consists of two stages of the same weights: pretraining **from scratch** over codec tokens → **fine-tuning** the same model on extended data and augmentation with a flat LR schedule.

Compute. The released model cost **≈63 GPU-hours on a single GPU**: 40.7 h of pretraining (through step 107k) plus 22.3 h of fine-tuning on extended data and the new augmentation (through step **160k**). That is the *entire* compute behind the shipped weights. The exploratory runs of earlier architectures are a separate story and are not counted here: they produced no checkpoint, only lessons (Sec. 3).

Scale — and an honest appraisal of it. By the standards of the field this is a **small** budget on both axes. We have **681.5 h** of data, and only **28%** of it is native wideband; the remaining 72% are upsampled 24 kHz targets, where content above 12 kHz *physically does not exist* and the model learns to synthesize it only indirectly. For comparison, RE-USE [5] trains on ≈2565 h — nearly four times more. We did **not** scale either the data or the compute to their limits: the results shown are what the architecture and the augmentation recipe yield on a small budget. The ceiling reached here is therefore not the ceiling of the method — raising the share of native wideband is the most direct lever, and it is one we did not pull.

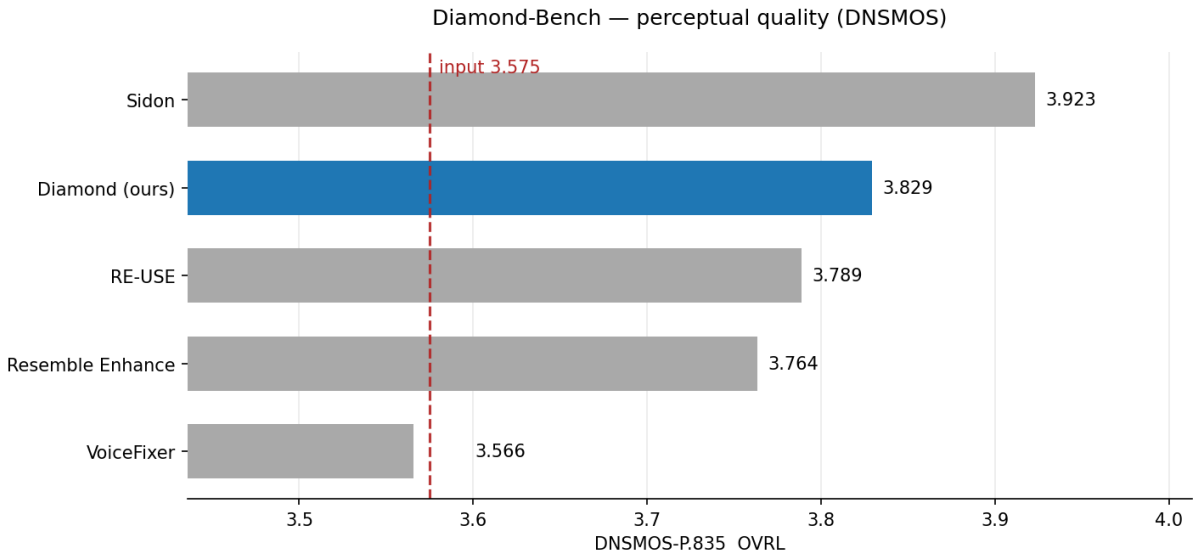


Figure 6: DNSMOS-P.835 OVRL against open baselines. Diamond is on par with the strongest current systems.

5. Inference

Long AR decoding accumulates exposure bias, so inference is chunked and guarded: chunk 2.5 s / overlap 0.4 s / warmup 1.0 s / tail-pad 1.0 s, repetition penalty ρ=1.3, silence-collapse recovery, and peak normalization to 0.99. Chunks shorter than the training crop keep the decoder within its trained horizon and are cross-faded together. Before every argmax we apply a **repetition penalty** (CTRL-style [17]) over a window $\mathcal{R}$ of recent tokens:

$$z_v \leftarrow \begin{cases} z_v/\rho, & z_v > 0, \\ z_v \cdot \rho, & z_v \le 0, \end{cases} \qquad v \in \mathcal{R}, \tag{11}$$

which breaks AR repetition loops — the single largest gain in content preservation.

The quality $\leftrightarrow$ speed trade-off. Autoregression generates tokens sequentially (86 frames/s $\times$ depth) $\rightarrow$ slower than non-AR feature predictors, but it delivers high per-frame quality and crisp speech; the depth-transformer adds $\approx \times 1.5$ compute per frame but fixes the fine codebooks; the no-downsample encoder is heavier but preserves detail. At inference the cost is mitigated by a **KV cache** ($O(T)$) and **batching of independent chunks**; the serial nature of AR remains the ceiling. For offline restoration this is the right trade.

6. Evaluation

6.1. Data and metrics

Evaluation is on **Diamond-Bench**: **750** real bad recordings (degraded recordings with reference transcripts). All models are run on the same 750 clips in the same order. Two axes: **DNSMOS-P.835** [14] (`sig_bak_ovr.onnx`, raw windowed mean; SIG/BAK/OVRL) and **CER** (faster-whisper [29] `small.en` + jiwer [30]), CER $= (S + D + I)/N$ — a safeguard against hallucination. All numbers are on the raw scale and are comparable only within the tables of this report; the outputs of all models and the input are peak-normalized uniformly.

Table 3: Results (DNSMOS-P.835 + CER).

Model	OVRL	SIG	BAK	CER_{med}	CER_{μ}
Sidon [18]	3.923	4.129	4.416	0.016	0.120
Diamond (ours)	**3.829**	4.056	4.348	0.028	0.131
RE-USE [5]	3.789	4.003	4.344	0.014	0.128
Resemble Enhance [20]	3.764	3.994	4.301	0.027	0.140
UniSE [42]	3.752	4.014	4.261	0.027	0.131
VoiceFixer [6]	3.566	3.790	4.225	0.042	0.150
input (degraded)	3.575	3.890	4.082	0.014	0.124

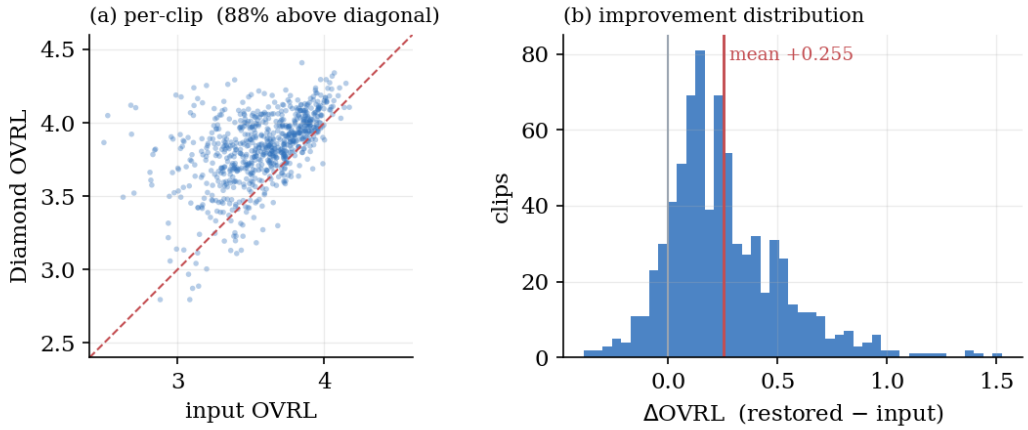


Figure 7: (a) Input vs restored OVRL per clip — points above the diagonal are improved (≈88%). (b) Distribution of ΔOVRL, mean +0.255.

6.2. Results

Relative to the input, Diamond improves **657/750 clips (≈88%)**, with a mean ΔOVRL = +0.255 (median +0.211).

Analysis. On perceptual quality Diamond ranks **second of six** (OVRL 3.829), behind only Sidon [18] and ahead of RE-USE [5], Resemble Enhance [20], UniSE [42], and VoiceFixer [6], with clear, intelligible restored speech (SIG 4.056). All of this while training **from scratch**: the systems that beat us stand on large pretrained feature extractors, and among restorers trained *without* a pretrained backbone Diamond is the best open one we know of.

On content preservation the picture inverts, and therein lies the main lesson of this work. Diamond's median CER is 0.028 — indistinguishable from UniSE (0.027) and Resemble Enhance (0.027), while RE-USE (0.014) and Sidon (0.016) lead. Both CER leaders are **non-autoregressive**: they regress features or spectra and do not *re-generate* the content at all. The one other *autoregressive* system in the table, UniSE [42], lands on exactly our mean CER (0.131) — by a different route (a pretrained WavLM [43] conditioning that anchors the content, and a token stream five times shorter than ours), which suggests the tail is a property of the paradigm rather than of our particular recipe. We do generate it, token by token, and over a long chain the error compounds (exposure bias, Sec. 5). The median is low — the typical clip is restored accurately; the higher mean (0.131) comes from a **thin tail** of recordings where generation derails. This is not a tuning defect but the **structural price of autoregression**, and it is precisely where the next work goes.

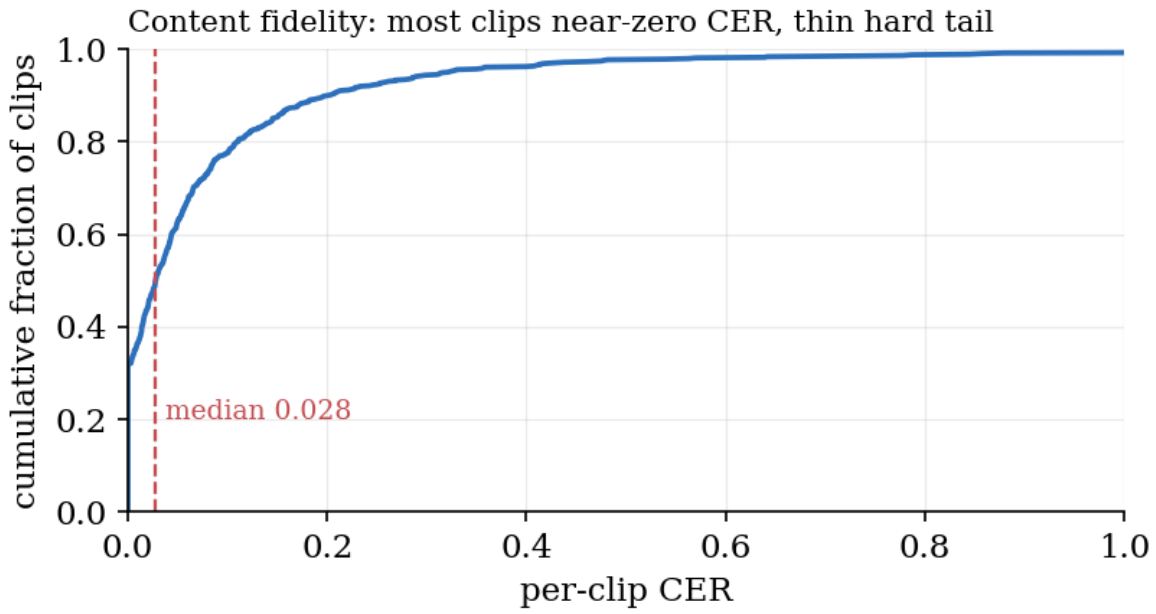


Figure 8: Cumulative distribution of CER: the typical clip is transcribed almost perfectly (median 0.028 ≈ the input's error), while the higher mean comes from a thin tail.

7. Conclusion

Diamond is a sequence-to-sequence speech-restoration model over neural audio codec tokens whose read-out is, to the best of our knowledge, the first two-transformer RQ decoder used for this task: a frame-level time-transformer plus a frame-local depth-transformer along the RVQ axis, in place of parallel heads [41]. Trained **from scratch** on 681.5 h (28% native wideband), it reaches **second place of five** on perceptual quality (+0.255 OVRL, 657 of 750 clips improved), trailing only Sidon, and remains **the best open restorer without a pretrained backbone that we know of**. The price is content: on CER we place third-to-fourth, and both systems ahead of us are non-autoregressive. This is the **structural price of generation** over regression, and we measure it openly rather than hide it. The engineering contribution is the diagnosis and elimination of the instabilities of deep training over codec tokens, with mechanistic explanations grounded in the literature. The main remaining weakness is the CER tail of autoregression.

A. Training configuration (as used)

Values are taken from the config of the released checkpoint (fine-tuning, step 160000).
Methods: pretraining from scratch → fine-tuning; teacher forcing (*no* scheduled sampling); weighted CE + label smoothing + z-loss; stabilization via norm_in + LayerScale + QK-Norm + gradient clipping; online DSP augmentation + noise/RIR transfer; variable crop + start-at-zero.

Table 4: Optimization.

Parameter	Value
Optimizer	AdamW, $\beta = (0.9, 0.95)$
lr	5×10^{-5}
lr schedule	flat; warmup 500; min-lr-ratio 0.1
Weight decay	0.01 (excl. bias/norm/embedding)
Grad clip (norm)	1.0
Precision	bf16-mixed
Batch	$8 \times$ grad-accum $4 =$ eff. 32
Steps	160 000
Scheduled sampling	disabled

Table 5: Data, crop, and augmentation (per-sample probability).

Parameter	Value
Crop	random $[2.5, 10.0]$ s; start-at-zero 0.2
Noise transfer	0.15; SNR $[0, 20]$ dB
RIR (reverberation)	0.15
Packet loss	0.15
Hard clipping	0.10
DSP augmentor	jitter 0.4; strategy-mix $[0.6, 0.25, 0.15]$

B. Notation

Symbol	Meaning
x	degraded input signal (waveform)
$\mathbf{X} \in [-1,1]^{80\times T}$	log-mel features of the input, 80 channels, T frames
T	number of frames (≈86 Hz)
$f_{\text{frame}}, f_{\text{DAC}}$	frame rate of the mel features and of the DAC tokens
$\mathbf{c} \in \{0..1023\}^{9\times T}$	target DAC RVQ codes; $\hat{\mathbf{c}}$ — the same targets
$c_{t,k}$	code of codebook k at frame t; $c_{t,<k}$ — codebooks $0..k{-}1$ of the same frame
$k \in \{0,\dots,8\}$	codebook index (RVQ depth), c_0 coarse, c_8 fine
$\mathbf{H}$	encoder output (cross-attention context)
$h_t \in \mathbb{R}^{512}$	hidden vector of frame t (time-transformer output)
$W_k : \mathbb{R}^{512} \to \mathbb{R}^{1024}$	output linear head of codebook k
$\hat{p}_{t,k}$	predicted distribution over codebook k
Ω	set of valid (non-pad) positions (t,k)
w_k	weight of codebook k in the loss
ϵ	label smoothing, $\epsilon = 0.1$
$\mathcal{L}_{\text{CE}}, \mathcal{L}_z, \mathcal{L}$	cross-entropy, z-loss, total loss
z, z_v	logits (over the vocabulary v); λ_z — z-loss weight
γ	learnable RMSNorm gain
λ	diagonal LayerScale scale (per-branch)
$\alpha^{(l)}, b$	per-layer RoPE scale ($\in [0.1, 2]$), base $b = 10000$
$\rho, \mathcal{R}$	repetition penalty ($\rho = 1.3$) and the window of recent tokens
S, D, I, N	substitutions/deletions/insertions and the number of reference characters (CER)

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